

## Reviewer: 1

### Comments to the Author

This paper attempts to answer the question: "Can basic sociodemographic characteristics like race and class predict individual-level lifespan?" The competing hypothesis set out in the paper is that instead of sociodemographic characteristics explaining individual longevity, mortality is a stochastic process that is unpredictable. The study uses data exclusively on men from the 1940 U.S. Census linked to Social Security Administration death records and employs a machine learning approach to try to predict individual survival from both a cohort and period perspective. The paper concludes that longevity is not explained by sociodemographic variables and that demographers therefore need to "account for stochastic processes". While the dataset seemed potentially interesting and the use of machine learning methods might be novel in this area of research, the paper on the whole is poorly thought out and its conclusions overstep the evidence, which is minimal. I have provided some comments and suggestions for future revisions or work in this area.

### Areas for improvement:

#### 1. Is this correct?

The study uses a set of machine learning algorithms to train a model designed to predict individual-level mortality/age at death in a holdout sample using information on early adulthood educational attainment, race, occupation, homeownership, etc. The findings indicate that the trained models have somewhat low predictive accuracy. The authors thus claim that these variables are not important for informing policy, since targeted individual health interventions would only be helpful if the models could predict individual mortality on the basis of the aforementioned variables. In specific, the authors write that: "...a stronger understanding of the predictability of longevity can help policymakers assess the feasibility of targeted, individualized health interventions."

This seems like flawed logic. Is the suggestion here that if race and class are not found to be highly predictive of individual longevity (as claimed in this paper), policymakers should assume that policies designed to improve mortality for racial minorities and poor people would not be feasible? Or that "individualized health interventions" would not be feasible for racial minorities and poor people? Either conclusion is at odds with decades of mortality research. The authors are comfortable making quite a sweeping conclusion on the basis of very little evidence. Let's take account of what we do know from this

study. We know that the specific models trained on the specific data applied to men born over 100 years ago do not appear to be provide accurate mortality predictions at the individual level. That is more likely to mean the models or data are bad, not necessarily that race and education are unimportant for individual mortality outcomes. If the authors do wish to push their claim that is in contrast with thousands of published works, they would need a mountain of evidence, not a molehill.

## 2. Is this important?

It's not clear why demographers should care about predicting individual mortality outcomes. Demography is the study of populations, not individuals, and that is because demographers understand that group-level processes and institutions matter for demographic outcomes. But even if we accept the premise, we're left with the question of what to do with this information. The authors write that demographers need to "account for stochastic processes." I think the authors are using the term "stochastic" in a non-standard way (or at least it is unclear what they mean when they call for more attention to stochasticity). A stochastic process is simply a random process. Mortality is indeed a stochastic process, so it is unclear why the authors write that researchers need to account for the stochastic nature of mortality. Any study of mortality is accounting for the stochastic nature of mortality. Do the authors perhaps instead mean that traditional demographic variables collected in almost all surveys are not enough, and that mortality researchers instead need to consider additional or more refined explanations? If that is the case, then I would point out that there have been thousands of published studies that have attempted to add the literature by drilling down more deeply or adding new variables or contextual information in trying to explain why mortality varies. This brings us to the next point.

## 3. Is the study well-written?

No, the study is not well-written. I do not mean that the writing itself is of poor quality, which is not the case, but instead I mean that the argumentation is poor. The authors have paid almost zero attention to the vast literature on racial and socioeconomic inequalities in mortality. They seem to think that their paper should be a wake-up call to mortality scholars to explore variables beyond just race and class, but they have ignored the many studies that have examined how mortality varies as a function of health behaviors, biomarkers, geographic characteristics, historical discrimination, access to healthcare, exposure to racism, genetic markers, mental health, cognitive functioning, psychosocial characteristics, etc. I do not know anyone who is making the claim that race and class are the only factors, or even the primary factors, that one should consider when studying mortality.

The paper's avoidance of the broader literature shows up in the referencing. The study not only does not cite many of the types of studies I mention above, but it also has strange citation patterns elsewhere. In many places throughout the paper, the authors cite articles that are not really relevant to their points. For example, they cite Ruggles 2020 in order to support the claim that large-scale microdata has only recently become available. Census microdata has been available to researchers for decades, well before IPUMS helpfully assembled the data in a more streamlined format. On the same page, the authors cite four papers as evidence that demographers are increasingly interested in predicting individual-level longevity, but only one of those papers is by demographers and it is not yet published. Another of the papers isn't even about mortality. When discussing racial inequalities in mortality, the authors mention only two studies, neither of which attempts to explain contemporary racial inequalities in mortality. When writing about SES and mortality, the study ignores most of the key figures and recent findings in that area. For both race and class, the authors do not discuss the bevy of articles identifying the key roles of drug use, smoking, cardiovascular disease risk, residential segregation, or any other contextual or individual-level variable in driving mortality variation. The authors should put in a much stronger effort to accurately reflect the state of the literature and our understanding of mortality variation today. If the authors are not mortality experts, they should consult one.

#### 4. Are the data and methods adequality described and justified?

It could be better. The authors mention some of the statistical/ML approaches that they use, but they do not describe them in nearly enough detail. This section should describe the methods in a way that justifies their use and also is accessible to demographers working on mortality. Given the relatively different methodological toolkit being employed compared to most demographic studies, it would be sensible to also have an extended appendix that describes these methods in much more detail than is currently given. Most importantly, the authors should justify the specific algorithms they're using. Why should readers think these algorithms are any good at prediction in general? Why should they expect the algorithms will be good at mortality prediction?

Why is it important to use the CenSoc-DMF linkage for this study? There are many other higher quality datasets out there that have information on education, income, housing, etc. and are linked to mortality follow-up, and many of them have sample sizes that rival what was used in this study. One added benefit of these other datasets is that they capture 21st century mortality instead of focusing almost exclusively on mortality from 50+ years ago. Another benefit is that they have a much better linkage percentage than the 17% in CenSoc-DMF. Because of the data issues in this paper, it's difficult to assess how much of the finding is due to data quality versus something else.

## Reviewer: 2

### General Comment

The topic of this paper is interesting, but I don't believe this paper belongs in this journal, because it misrepresents what mortality demography is about. Mortality rates are highly predictable, with an  $R^2$  greater than .98. Individual mortality of course has a lot of stochasticity to it. Throughout most of the life span, mortality risk is very low. Even at the older ages, mortality rates are not particularly high until after age 80 or so. So, if one wished to maximize  $R^2$ , one would simply predict that no one dies. This is the same problem with logit models in cases in which the dichotomous outcome is not in the neighborhood of 50%. It simply defies reason that one would possibly expect that a measure like sex, with effectively two values, could even possibly facilitate individual level prediction when there are hundreds of millions in the population and only two values in the variable! Couple that with low rates, and it is absurd to think one could predict individual mortality. Think about this: suppose the mortality probability within a covariate category is 2% over a 10 year period, while it is double this (4%) in another category. The relative risk ratio between the two categories is 2, and that's a substantial between-group effect. Yet, if you had 100 persons in the first group, you would predict two deaths, but with no additional information, you have no basis for expecting you'd predict the correct two—there'd be less than .03% chance of selecting the two correctly, even if the prediction that 2 would die was exactly correct. I have shown this in one of my own, older papers. Obviously, the authors are using more covariates, but the total number of cells cannot be even close to the number of persons in the sample, and so the logic holds.

So, the authors have effectively set up a strawman that the goal of mortality demography is individual prediction and then they've knocked it down by showing that imprecise covariates can't identify who, within a broad covariate category/cell, dies within an arbitrary time window, despite the fact that this is mathematically obvious with generally low mortality rates at most ages. In fact, I'd argue that the authors would reach a different conclusion if they changed the approach to predicting who SURVIVES a 10 year window—that makes it fairly obvious that the lack of predictability is the consequence of predicting rare events.

If demography's focus were on individual level mortality, demographers would not use imprecise, broadly measured covariates to predict it. Instead, family history, individual

past health history, and a larger, more refined set of covariates would be used. The authors indicate that other research has done this to varying extents, but most of the research cited is not by mortality demographers! This is because individual prediction is not the goal of mortality demography. Instead, demography's focus, especially when imprecise sociodemographic covariates are used, is simply to predict aggregate mortality, which certainly has large scale policy implications. Obviously, greater focus should be on the extent of heterogeneity within broad sociodemographic categories, but that is precisely a contemporary focus of mortality demography, as the authors discuss.

All that said, I am sympathetic to the idea that lay people—and even scholars—often misunderstand what is meant when we say something like “life expectancy for white men at age 50 is 31 years, give or take .5 years.” What is meant is that the average years remaining for members of the group is between 30.5 and 31.5 years—it says nothing directly about specific individuals within the group. There is ultimately no way to eliminate/explain within-group variability, because the event (death) is a Bernoulli distributed random variable that is iid within groups by assumption. That means the units within cells are exchangeable, meaning there is no basis for predicting that one person experiences the event while another doesn't. Put another way, the  $p$  parameter of the Bernoulli represents the expectation in the aggregate. No individual can have a death value of, say .07. But, that  $p$  indicates 7/100 exchangeable individuals will die, and models in mortality demography are quite good at accurately predicting that.

Some specific comments

p.2. first paragraph. This is a false dichotomy—mortality can (and does) have both a structured and stochastic component. The seemingly large stochasticity at the individual level is an artifact of low rates and some basic probability theory.

p.2., line 40. I disagree with the premise that the predictability of individual mortality has scientific and policy implications. Mortality RATES are highly predictable and are the real basis for policy, like making Social Security and Medicare solvent. Further, even if there is a lot of individual variability in mortality, the fact that there may be clear patterns by race, class, etc., suggest that the AVERAGE person will live longer if race, class, etc., disparities are reduced or eliminated. The prediction of individual mortality using these broad covariates has no real policy implication. For individuals, family history and individual health history would be relevant, and you don't have that information here, and most studies in demography do not have representative data for those kinds of covariates.

p. 8, line 42. The correct word here is “precise” not “accurate.” I can absolutely predict with great accuracy that everyone on the planet alive today will die within 120 years (barring no unforeseen breakthrough in immortality research), but it isn’t a very precise estimate.

p. 11, line 6. Again, the goal of demography is not to predict individual mortality. These theories are not necessarily intended to predict individual mortality but patterns of mortality among groups. At that level, the theories are supported by very strong findings: aggregate mortality is very predictable.